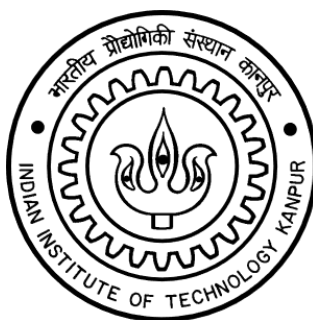


LABORATORY MANUAL

TA211

MANUFACTURING PROCESSES-I

2025-2026 Semester-I



Indian Institute of Technology, Kanpur

Department of Materials Science and Engineering

<u>COURSE INSTRUCTOR</u>	<u>LAB INCHARGE</u>
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Laboratory	Monday ,Wednesday and Friday (As per one of the assigned days of your section) 02:00 pm to 05:00 pm Venue: Engineering Metallurgy Lab DJ Building 101 H.
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Experiment 1 (E1)

Sand Moulding & Casting and Forging

(A) Theory

- **Sand Moulding and Casting**

Introduction to Casting Process: Casting is one of the oldest and the most popular processes of converting materials into useful shapes. The casting process is primarily used for shaping metallic materials; although it can be adopted for shaping other materials, such as ceramic, polymeric and glassy materials. In casting, a solid is melted, treated to the proper temperature and then poured into a cavity called mould, whose shape is similar to final desired shape in which molten metal spreads completely before solidification starts. Simple as well as complex shapes can be made from any metal that can be melted. The resulting product can have virtually any configuration the designer desires.

Casting products range in size from a fraction of centimeter and fraction of kilogram to over 10 meters and many tons. Moreover, casting has marked advantages in the production of complex shapes, of parts having hollow sections or internal cavities, of parts that contain irregular curved surfaces and of parts made from metals which are difficult to machine.

Several casting processes have been developed to suit economic production of cast products with desired mechanical properties, dimensional accuracy, surface finish etc. These various processes differ primarily in mould material (whether sand, metal or other material) and pouring method (gravity, pressure or vacuum). All the processes share the requirement that the material solidifies in a manner that would avoid potential defects, such as shrinkage voids, gas porosity and trapped inclusions. The casting process involves three necessary steps, i.e., mould making, melting and pouring of melt into the mould cavity, and removal and finishing of cast product after complete solidification. One of the significant classifications of casting is based on whether the mould is used again or it is prepared fresh every time. Sand casting is an example of an expendable mould process or where the mould is broken after every casting to remove the component. In this lab, students will practice this particular process by making the mould and then pouring the liquid metal into the mould to form a final component.

Sand Casting Processes: Sand is one of the cheaper, moderately refractory materials and hence is commonly used for making moulds. Sand contains grains of silica (SiO_2) along with a

few impurities. For mould making, sand is mixed with a binder material such as clay, molasses, oil, resin etc.

✓ **Green Sand Moulding**

In the green sand moulding process, clay (a silicate material) along with water (to activate clay) is used as a binder. The mould-making mainly consists of preparing a cavity having the same shape as the part to be cast. There are many ways to obtain such a cavity or mould, and in this demonstration, you will learn to make it using wooden 'pattern' and 'green-sand' as the mould material.

A pattern is a reusable form having approximately the same shape and size as the part to be cast. A pattern can be made out of wood, metal or plastic; wood is the most common material. Green sand refers to an intimate mixture of sand (usually river sand), bentonite clay (3-7% by weight of sand, to provide bonding or adhesion between sand grains), and water (3-6% by weight of sand, necessary to activate the bonding action of the clay).

Mixing the above ingredients in a sand muller provides an intimate mixing action. In practice, a significant part of this sand mixture consists of 'return sand', i.e. the reusable portion of the sand left after the solidified metal casting has been removed from the mould. Moulding flasks are rectangular frames with open ends, which serve as containers in which the mould is prepared. Usually, a pair of flasks are used; the upper flask is referred to as 'Cope' and the lower one as 'Drag' (Fig. 1). A riddle is a relatively coarse sieve which is used to break the lumps in the green sand and its aeration.

Sometimes the cast itself must have a hole or cavity in or on it. In that case, the liquid metal must be prevented from filling certain portions of the mould. A 'core' is used to block off portions of the mould from being filled by the liquid metal. A core is normally made using sand with a suitable binder like molasses. The core is prepared by filling the core-box with core sand to get the desired shape and the baking this sand core in an oven at a suitable temperature. During mould making, a suitable 'gating system' and a riser' is also provided. The gating system is the network of channels used to deliver the molten metal from outside the mould into the mould cavity. Various components of the gating system are pouring cup, sprue, runners and gates. Riser or feeder head is a small cavity attached to the casting cavity, and the liquid metal of the riser serves to compensate the shrinkage in the casting during solidification and ideally should be last to freeze. Fig. 1 shows the various parts of a typical sand mould. Several hand tools, such as

rammer, trowel, sprue pin, draw spike, slick, vent wire, gate cutter, strike-off bar etc. are used as aids in making a mould.

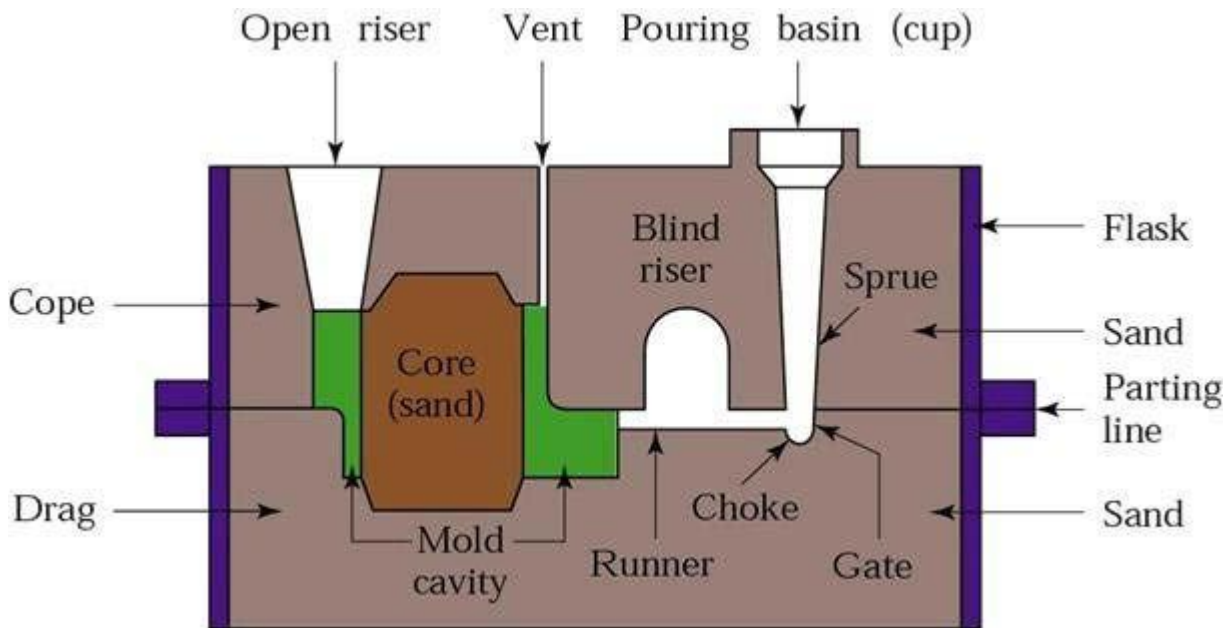


Fig. 1.1. Schematic of a typical two-part sand mould, indicating various mould components and terminology.

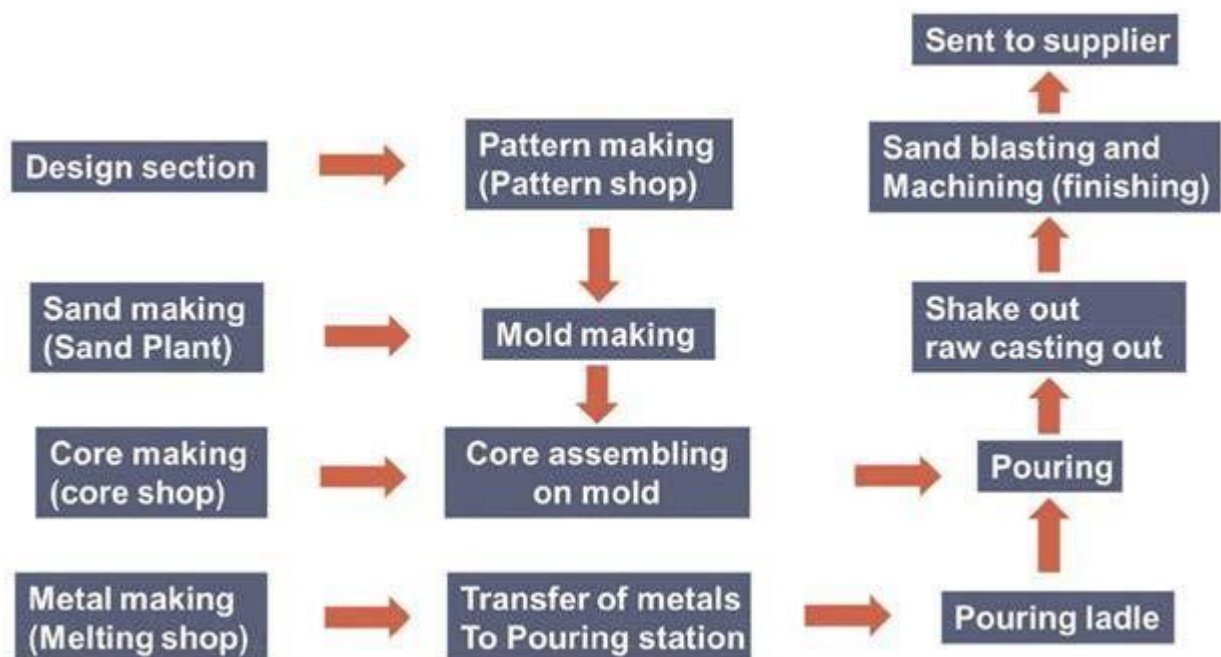


Fig. 1.2. Process of casting, including pre-processing and post-processing operations.

✓ **Melting and Pouring of Metals**

Next important step is the melting of metal. A melting process must be capable of providing molten metal not only at the proper temperature, greater than melting point, but also in the desired quantity, with acceptable quality, and within a reasonable cost. In order to transfer the metal from the furnace into the moulds, some pouring device, or ladle, must be used. The primary considerations are to maintain the metal at the proper temperature for pouring and to ensure that only quality metal will get into the moulds (without slag or oxide). The operations involved in the melting of metal in induction furnace and pouring of liquid metal into the mould cavity will be shown in the lab.

✓ **Removal and Finishing of Castings**

After complete solidification, the cast is removed from the mould. Most casts require some cleaning and finishing operations, such as removal of cores, removal of gates and risers, removal of fins and flash, cleaning of surfaces, etc.

• **Forging Process**

Introduction to Forging Process: Metal forging is a process of shaping and forming metal through the application of heat and pressure. It involves heating the metal to a specific temperature and then applying force to shape it into the desired form. Open die forging uses dies that leave the metal open, but closed die forging encases the metal, allowing for more precise forged shapes. During forging, several defects can occur, including 'Barreling' and 'Buckling'. Barreling is when the sample starts to form a barrel shape as it's being compressed in an open die forging process. Buckling is the sudden change in shape (deformation) of a structural component under load, such as the bowing of a column under compression.

(B) Exercise

The exercise will consist of moulding & casting (Part-1) along with the demonstration of aluminum forging (Part-2).

Part-1: Mould making & casting

Objectives

1. To prepare a green sand mould from wooden pattern.
2. To melt and pour Aluminum metal into the mould.
3. To prepare a pattern for a given object for lost foam casting (demonstration).
4. To prepare a molasses sand mold from the prepared thermocol pattern (demonstration).

Tools and Equipment: Wooden pattern, core box, molding box, screw, spoke lifter, vent wire, rammer, leveler, sand muller, core, baking oven, melting furnace, pouring ladle, pyrometer, hacksaw, etc.

Materials: Thermocol, River sand, bentonite clay, coal dust, molasses, aluminum, flux, etc.

Procedure of Green Sand Casting

- ✓ **Mixing:** Green sand and molasses sand will be prepared by mixing the ingredients in sand Muller (Fig. 1.3).



Fig. 1.3. Sand Muller for green sand preparation.

✓ **Mould making**

1. Place the drag part of the molding flask and fill molding green sand.
2. Place the wooden pattern at the center of the drag (flask)
3. Pack the sand carefully around the pattern. Heap more molding sand in the drag and ram with rammer carefully.
4. Place the other half of the pattern over the pattern (already placed on the drag) matching the guide pins. Then, place the gating system with sprue and riser in proper positions.
5. Repeat step 3 to complete the mould box. Remove the sprue and riser pins and make a pouring basin.

- Open the mould and remove the wooden pattern. Now, make runner of the gating system. Place the core and then close cope and drag.

Overall process is shown in Fig. 1.4.

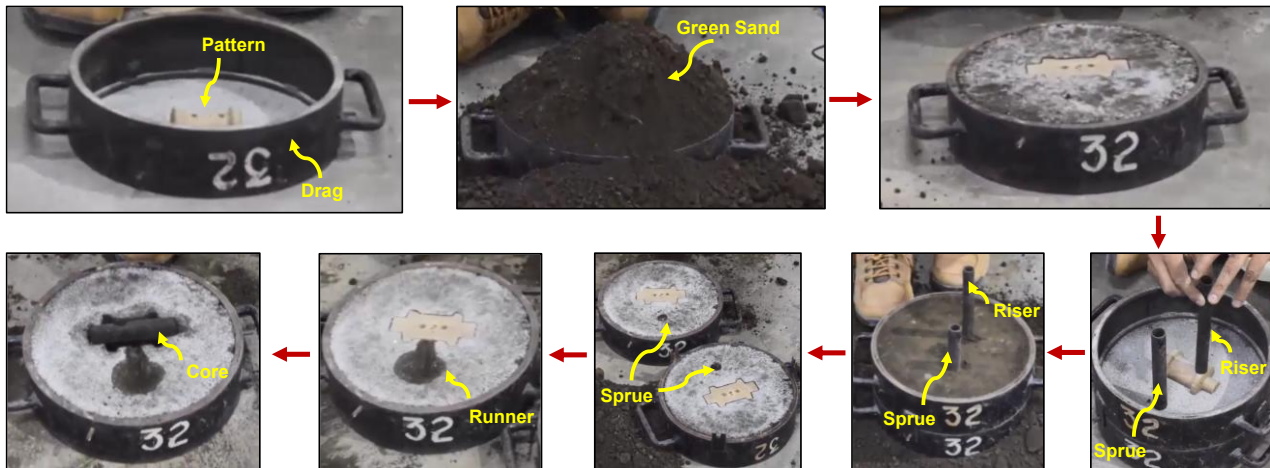


Fig. 1.4. Sequence of mould making process.

✓ Melting and Pouring

- Melt the metal in the furnace (Fig. 1.5a). Use appropriate fluxes at proper stages to prevent the oxidation of the metal. Measure metal temperature from time to time.
- Pour the molten metal into the pouring ladle kept at a higher temperature (say 100°C higher) than the pouring temperature. As soon as the desired pouring temperature is achieved, pour the liquid metal into the mould in a steady stream with ladle close to the pouring basin of the mould. Do not allow any dross or slag from the surface to go in the ladle.
- Allow sufficient time for the metal to solidify in the mould.

✓ Removal of cast from mould

- After the melt has solidified, break the mould carefully and remove the cast (Fig. 1.5b).
- Cut-off the riser and gating system from the cast and remove sand (if any).
- Inspect the cast visually and record surface and dimensional defects (if any).

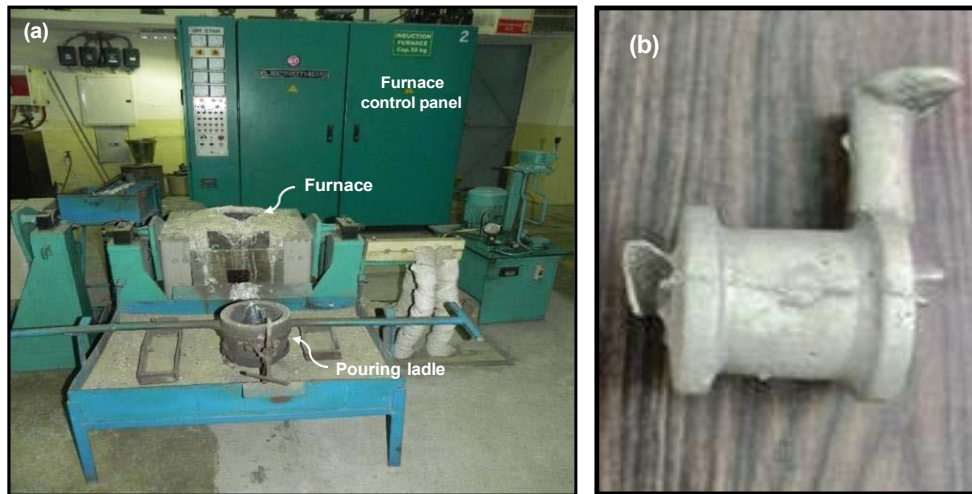


Fig. 1.5. (a) Furnace to melt metal and pouring ladle and (b) cast.

Demonstration of Lost Foam Casting using Molasses Sand

Lost foam casting process will be demonstrated. The process of mixing and moulding/casting/Melting & Pouring are same as that of greens sand casting. However, the pattern will be made of thermocol (Fig. 1.6). The final product after casting is shown in Fig.



Fig. 1.6. Foam pattern and the corresponding cast object.

Part-2: Forging

Objectives: Demonstration of forging of aluminium using press forging

Tools and Equipment: Vernier caliper, Forging machine

Materials: Aluminium

Procedure: Take aluminium cylinders (one large and one small height) and compress those using forging machine (Fig. 1.7a). The forging process is shown in Fig. 1.7b. The samples showing barreling and buckling during forging (Fig. 1.7c).

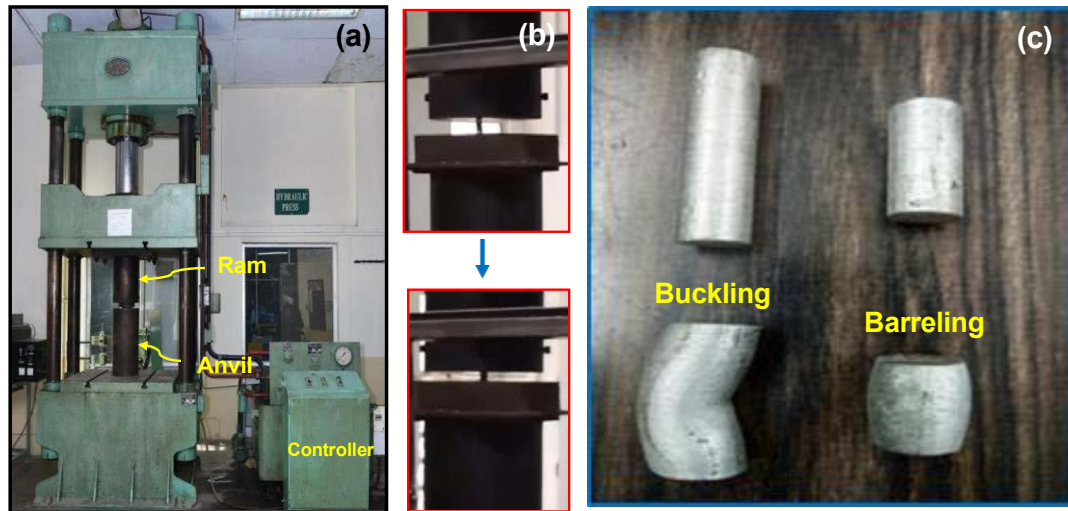


Fig. 1.7. (a) Forging machine, (b) Forging of cylinder, and (c) Buckling and barreling during forging

Experiment 2 (E2)

Sheet Metal Forming

(A) Theory

Introduction to Sheet Metal Forming: Products made of sheet metals are all around us. They include a very wide range of consumer and industrial products, such as beverage cans, cookware, file cabinets, metal desks, appliances, car bodies. The term press working or press forming is used commonly in industry to describe general sheet-forming operations, because they typically are performed on presses.

Many products are manufactured from sheet metal involving a combination of processes, such as shearing, bending, deep drawing, spinning etc. (Fig. 2.1). These processes are characterized by localized deformation and configuration changes and are together called “Sheet Metal Forming”. These processes do not result in a bulk shape change of the parts but lead to configurational changes. Sheets have large surface area to volume ratio of starting metal which distinguishes these from bulk deformation. Sheet metal forming is often called ‘press working’ as presses are required to perform these operations. It is also called stamping, where usual tools involve punches and dies. Sheet metal operations can involve a combination of stresses, for example,

- Stretching of the metal (tensile stresses)
- Bending of the metal (tensile and compressive stresses)
- Cutting of the metal (shear stresses)

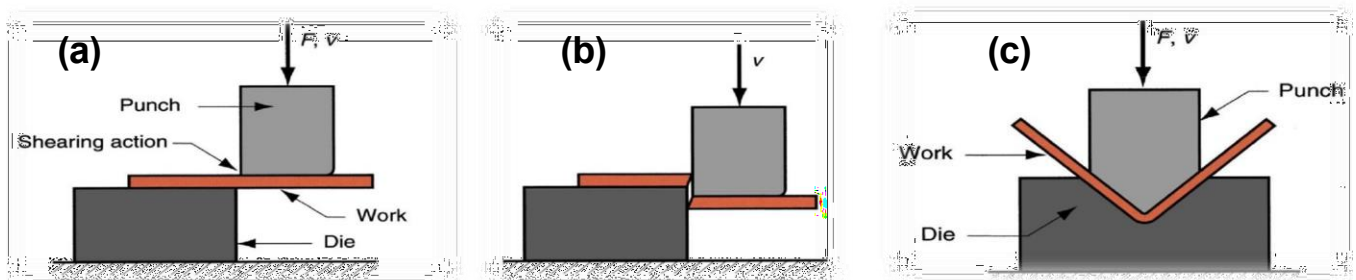


Fig. 2.1. Schematic of a few sheet metal working operations (a) initial configuration, (b) shearing, (c) bending

Shearing, bending and folding machines (for sheet metal) in the Engineering Metallurgy Lab are shown in Fig. 2.2.



Fig. 2.2. Some of the apparatus available in TA211 lab for sheet metal forming (a) shearing equipment, (b) bending equipment, (c) folding equipment.

(B) Exercise

The exercise will consist of making a sheet metal product (funnel) (Part - 1) and self-secured sheet metal joints (Part - 2).

Part-1: Sheet metal forming

Objectives: To fabricate a sheet metal product (Funnel).

Tools and Equipment: Mallet, hand snip, bench vice, grooving tool, scribe, scale, marker, light weight hammer, divider, power and manual shearing machines, three rolls bending machines, folding machine.

Materials: Galvanized iron (GI) sheet and rivets

The dimension of the funnel is shown in Fig. 2.3.

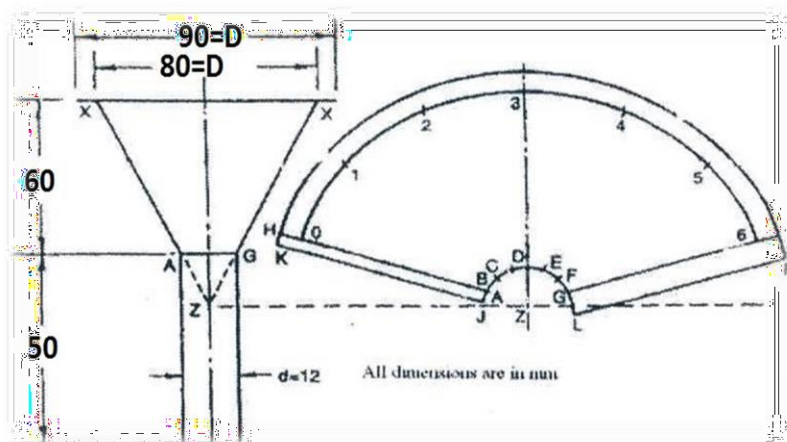


Fig. 2.3. Approximate dimensions of the funnel to be fabricated.

Procedure for making funnel:

- Draw the elevation on full scale.
- Complete the cone by extending the lines A and G
- Choose a point Z and draw curves with Z as a center, and ZA and ZX as a radius.
- Draw the vertical line Z3, meeting the internal curve at D, and external curve at '3'.
- Starting from D mark lengths DC, CB, BA, DE, EF and FG, each length equal to $\pi d/6$.
- Again, starting from point 3 mark length 3-2, 2-1, 1-0, 3-4, 4-5 and 5-6, each equal to $\pi d/6$. (D and d are major and minor diameters, respectively)
- Draw another curve with Z as a center and ZX+5 mm as a radius.
- Joint AO and G6 and extend it to cut the outer curve at points H and I, respectively.
- Provide a margin of 5 mm on one side and 10 mm on another side for joint.
- Cut out the required portion and form the conical portion.
- Make the bottom half of the funnel.



Fig. 2.4. The final form of funnel to be fabricated in the lab.

Part-2: Self-secured sheet metal joints (Fig. 2.5)

(a) Internal (Single) grooved joint

- Mark out portions of given sheets near edges to be joined with a marker.
- Fold the sheets at edges in the portion marked, first at right angles to the plane of the sheet and then at 180° to the plane.
- Insert one folded sheet into the other.
- Groove the seam using grooving die.

(b) Double grooved joint

- Fold sheets as per the instructions given.
- Cut a piece of a sheet (called strap) of required width.
- Strap width = (4 x size of marked edges) + (4 x thickness of sheet).
- Close the edges of the strap slightly.
- Slip the strap on the curved edges of the sheets.

(c) Knocked-up joint

- Fold one sheet and close the edges slightly.
- Bend one sheet to form a right angle bend.
- Slip the second sheet in the folded one.
- Close the right-angled sheet using a mallet.

(d) Rivet joints

- To make a rivet having both fixed and movable joints using GI Sheet.

(e) Hemisphere

- To make a hemisphere using GI Sheet with the help of anvil and ball peen hammer.

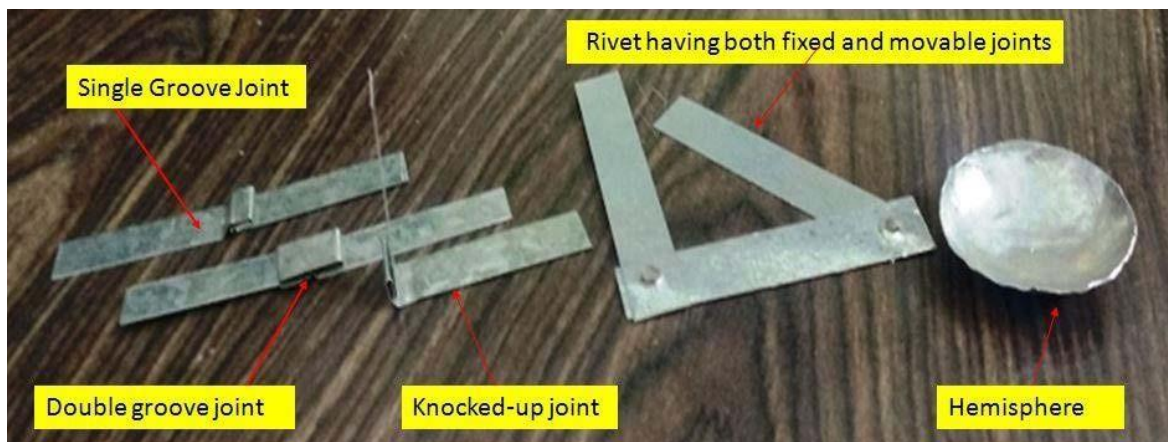


Fig. 2.5. Different joints and hemisphere.

Experiment 3 (E3)

Welding Processes, Powder Metallurgy and Plasma Cutting Process

(A) Theory

Introduction to Welding: Solid materials need to be joined together for various applications, such as industrial, commercial, domestic, art ware and other uses. Depending on the material and the application, different joining processes are adopted, such as mechanical (bolts, rivets, etc.), chemical (adhesive) or thermal (welding, brazing or soldering). Thermal processes are extensively used for joining metals.

Welding is a process in which two materials, usually metals, are permanently joined together by coalescence, resulting from temperature, pressure, and metallurgical conditions. The particular combination of temperature and pressure can range from high temperature with no pressure to high pressure. Thus, welding can be achieved under a wide variety of conditions and numerous welding processes have been developed and are routinely used in manufacturing.

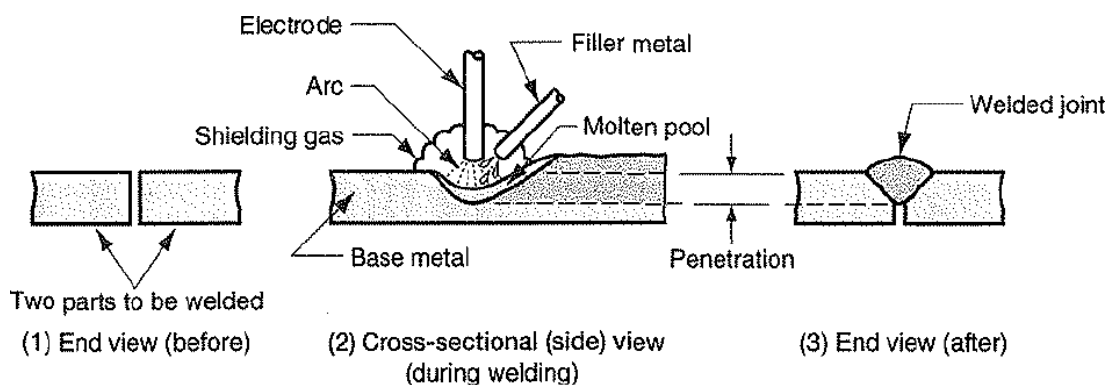


Fig. 3.1. Schematic of a welding process.

To obtain coalescence between two metals the following requirements need to be met: (1) perfectly smooth, flat or matching surfaces, (2) clean surfaces, i.e., free from oxides, absorbed gases, grease and other contaminants. Surface roughness is overcome by pressure or by melting two surfaces so that fusion occurs. Contaminants are removed by mechanical or chemical cleaning before welding or by causing sufficient metal flow along the interface so that they are removed away from the weld zone. In many processes, the contaminants are removed by fluxing agents. The production of quality welds requires (1) a satisfactory heat and/or pressure source, (2) a means of protecting or cleaning the metal, and (3) caution to avoid, or compensate for, harmful metallurgical effects.

Arc Welding

In this process, a joint is established by fusing the material near the region of a joint by means of an electric arc struck between the material to be joined and an electrode. A high current low voltage electric power supply generates an arc of intense heat, reaching a maximum temperature of approximately 5500°C. The electrode held externally may act as a filler rod (consumable electrode) or it is fed independently of the electrode (non-consumable electrode). Due to higher levels of heat input, joints in thicker materials can be obtained by the arc welding process. It is extensively used in a variety of structural applications. There are many types of the basic arc welding process, such as Manual Metal Arc Welding (MMAW), Metal Active Gas Welding (MAG), Gas Tungsten Arc Welding (GTAW), Submerged Arc Welding (SAW). Note that MAG is also known as Gas Metal Arc Welding (GMAW) and Metal Inert Gas Welding (MIG). MMAW is also known as Shielded Metal Arc Welding (SMAW). GTAW is also known as Tungsten Inert Gas (TIG) welding and W Inert Gas (WIG) welding. Also, it should be noted that GMAW, SMAW, SAW processes use consumable electrode, whereas GTAW uses non-consumable electrode. Schematic diagrams of SMAW and GMAW processes are shown in Fig. 3.2.

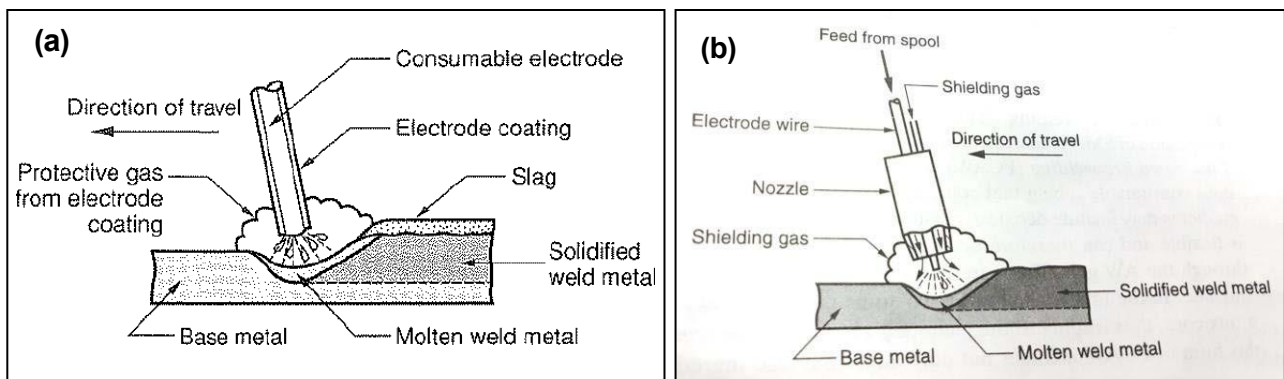


Fig. 3.2. Schematic of a (a) shielded metal arc welding process (SMAW) and (b) gas metal arc welding (GMAW).

Introduction to Powder Metallurgy: Powder metallurgical (P/M) processing of the metals and alloys plays an important role in manufacturing various engineering components for several applications. The conventional P/M technology for making components starts with metal powders as the raw material, which is pressed in dies of suitable shape to produce green compacts. These green compacts are subsequently, sintered at high temperature under protective atmosphere with following objective: (i) to develop proper bonds between higher to mechanically pressed powder particles comprising the green compacts and (ii) to reduce the porosity level in the sintered

compact so that it has still higher relative density. However, parts produced by the conventional P/M approach always contain some residual porosity. Non-conventional P/M technologies have been developed with the objective of completely eliminating the porosity from the manufactured part.

Metal powders are produced by various methods, such as (a) atomization, (b) solid-state reduction of metal oxides or other compounds, (c) chemical precipitation from organic or inorganic solutions, (d) electrolysis, (e) mechanical alloying etc. Each of these generic processes may have several specific variants for metal powder preparation. The shape, size and their distribution of as-synthesized metal powders depend on the specific method of production, variables associated with the process and their control. The tapping of a powder stock in the die and its behavior during compaction and sintering is substantially influenced by its size and shape characteristics. It can be mentioned here that the spherical powder particles with uniform size distribution is usually preferred from the point of view of good flowability during compaction as well as for good sinterability.

The density of loose powder is known as the apparent density. It is measured by allowing the loose powder to flow through the Hall flow meter. Very often, the loose powder is mechanically vibrated and/or tapped so that the particles in the loose mass further settle and occupy a smaller volume.

Introduction to Plasma Cutting: Plasma cutting, also referred to as plasma fusion cutting or plasma arc cutting, is a fabrication process which employs superheated, ionized gas funneled through a plasma torch to heat, melt and, ultimately, cut electrically conductive material into custom shapes and designs. This process is suitable for a wide range of metal materials, including structural steel, alloy steel, aluminum, and copper, and can cut a wide range of thicknesses. When the plasma jet hits the workpiece, recombination takes place and the gas reverts to its normal state, emitting intense heat as it does so. This heat melts the metal and the gas flow ejects it from the cut. Plasma gases are usually argon, argon/hydrogen or nitrogen. Plasma arc can cut a very wide range of electrically conductive alloys including plain carbon and stainless steels, aluminium and its alloys, nickel alloys and titanium. The method was originally developed to cut materials which could not be satisfactorily cut by the oxy-fuel process. Normally, the component or sheet to be cut remains stationary and the plasma torch is moved.

Introduction to Spot Welding: It is a form of resistance welding, where two or more sheets of metal are welded together without the use of any filler material. The process involves applying

pressure and heat to the weld area using shaped alloy copper electrodes which convey an electrical current through the weld pieces. The material melts, fusing the parts together at which point the current is turned off, pressure from the electrodes is maintained and the molten “nugget” solidifies to form the joint (Fig. 3.3).

The welding heat is generated by the electric current, which is transferred to the workpiece through copper alloy electrodes. Copper is used for the electrodes as it has a high thermal conductivity and low electrical resistance compared to most other metals, ensuring that the heat is generated preferentially in the work pieces rather than the electrodes. The amount of heat depends on the thermal conductivity and electrical resistance of the metal as well as the amount of time the current is applied. This heat can be expressed by the equation:

$Q = I^2Rt$, where ‘Q’ is heat energy, ‘I’ is current, ‘R’ is electrical resistance and ‘t’ is the time for which the current is applied.

Spot welding has applications in a number of industries, including automotive, aerospace, rail, metal furniture, electronics, medical, construction, etc. Given the ease with which spot welding can be automated when combined with robots and manipulation systems, it is the most common joining process in the construction of steel cars.

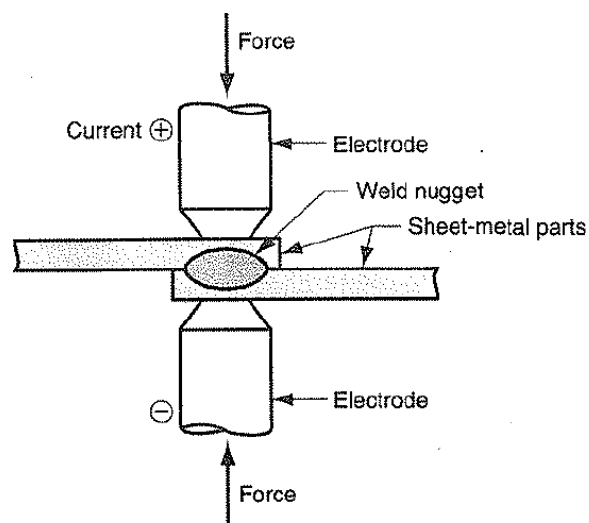


Fig. 3.3. Process of spot welding.

(B) Exercise

The exercise will consist of making a joint by arc welding process (Part - 1). Demonstration of P/M process, spot welding and plasma cutting process (Part - 2).

Part-1: Arc welding process

Objectives: To prepare a butt joint with mild steel flat using Metal Active Gas Welding (MAG) and Manual Metal Arc Welding (MMAW) technique.

Tools and Equipment: Welding unit, Protecting gas, Wire Brush, Tongs, etc.

Materials: Mild steel flats (100 x 25 x 5 mm), Consumable mild steel wire

Procedure

- Clean the mild steel flats with a wire brush.
- Arrange the flat pieces properly providing the gap for full penetration for butt joint (gap to be $\frac{1}{2}$ of the thickness of flats).
- Practice striking of arc, speed and arc length control.
- Set the welding current, voltage according to the type of metal to be joined. Strike the arc and make tacks at both ends to hold the metal pieces together during welding process.
- Lay beads along the joint maintaining proper speed and arc length (Speed 100-150 mm/min).
- Clean the welded zone and submit the job.

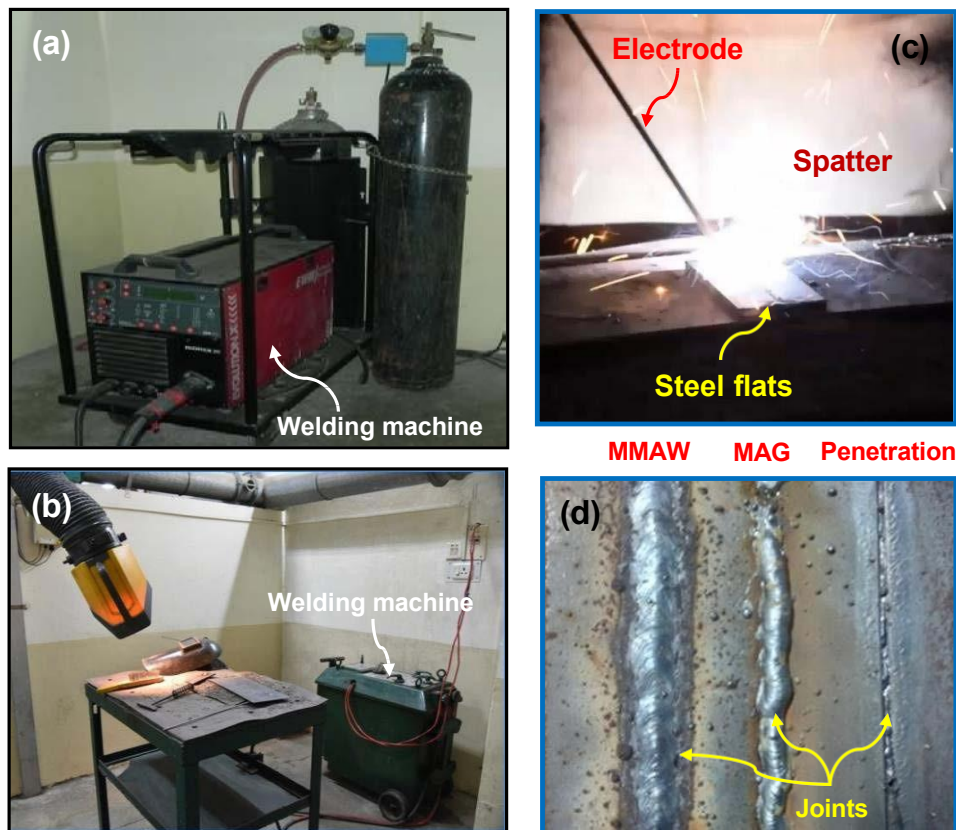


Fig. 3.4. Welding machines in TA211 lab: (a) MAG and (b) MMAW. (c) Welding process, (d) Joints using MMAW and MAG.

Part-2: Demonstration of powder metallurgy, spot welding and plasma cutting processes

Objectives:

1. To fabricate samples using P/M process.
2. To cut a 2 mm thick mild steel sheet using a plasma cutting process.
3. To create a joint between mild steel sheet (0.35 mm) using spot welding process.

Tools and Equipment: Spot welding machine, Hydraulic press machine, Furnace, Plasma cutting setup.

Materials: Copper powders and Mild steel sheets

Procedure:

1. Powder Metallurgy Process:

- Take copper powders and measure apparent density and tap density.
- Place the powders inside the die and prepare green pellet using hydraulic press.
- Place the green pellet inside the furnace to obtain the final sintered compact.

The overall process is shown in Fig. 3.5.

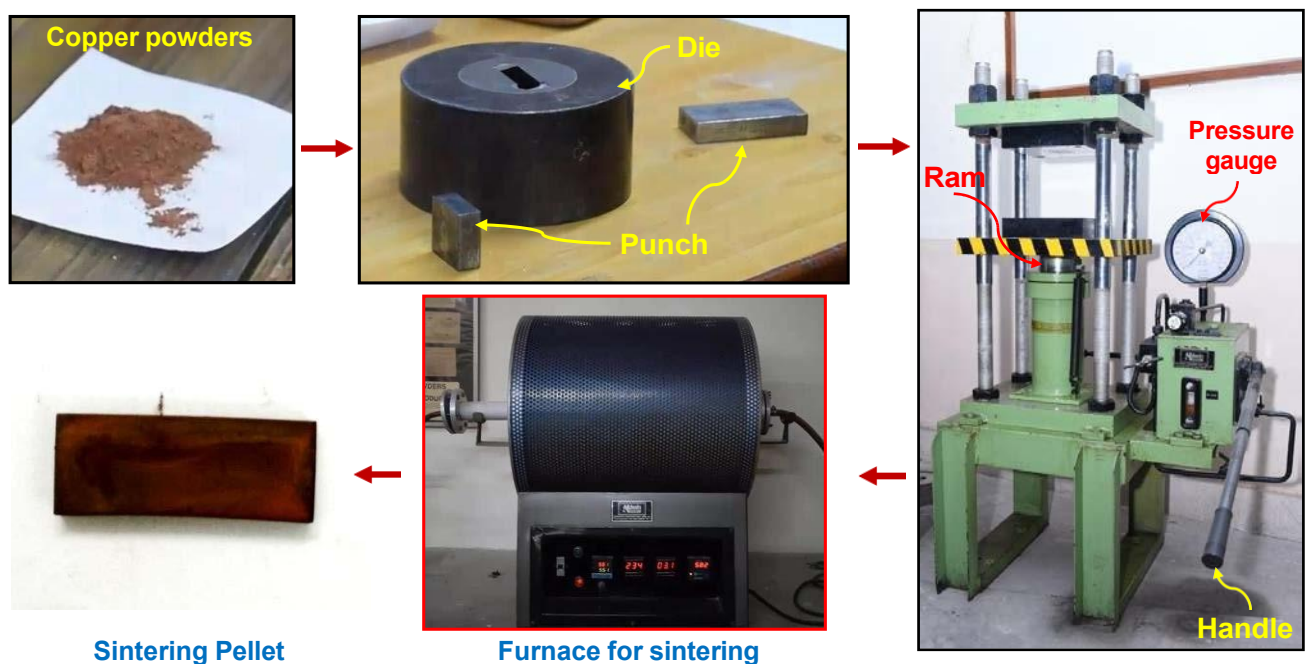


Fig. 3.5. Overall P/M process for fabricating samples.

2. Plasma Cutting Process:

Take 2 mm thick mild steel sheet and cut the sheet using plasma cutting machine. Fig. 3.6 shows the plasma cutting set-up along with process.

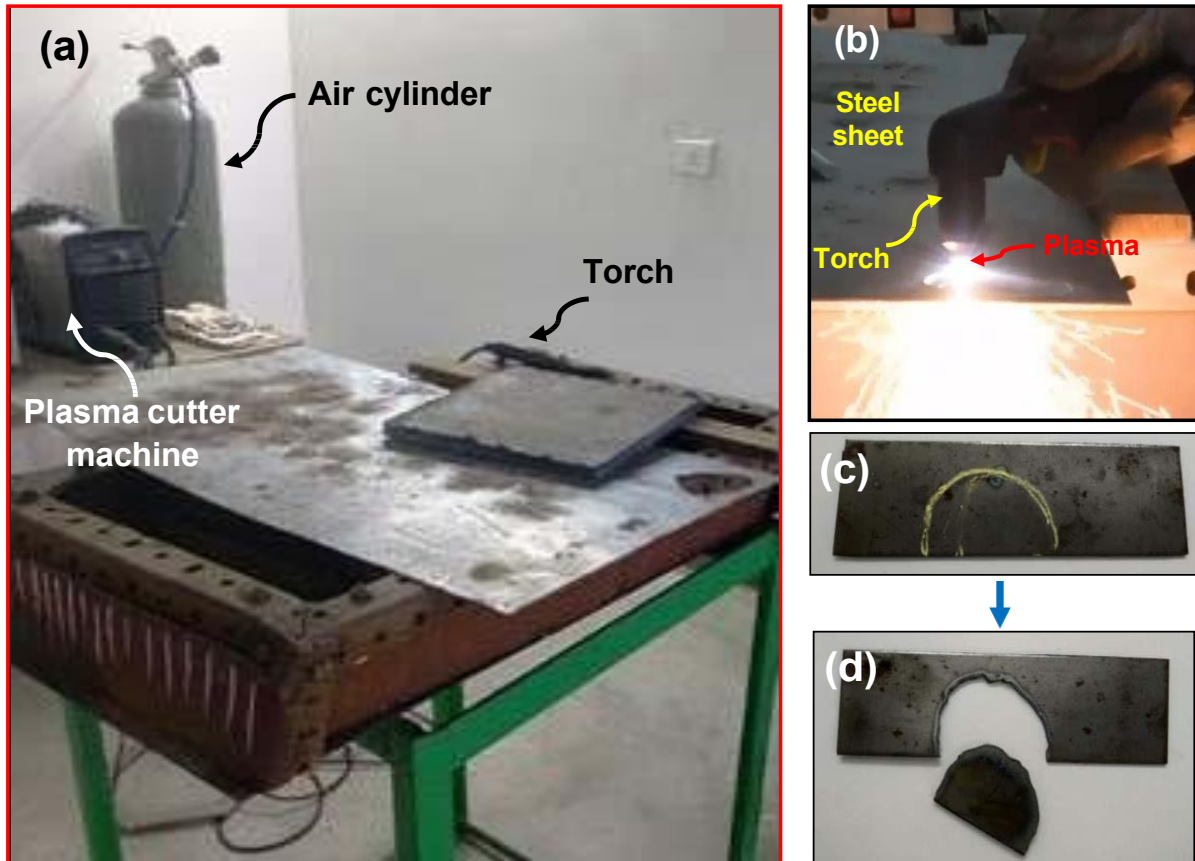


Fig. 3.6. Plasma cutting (a) setup and (b) process. (c) and (d) are sheets before and after plasma cutting.

3. Spot Welding Process:

Take two steel sheets (0.35 mm thick each) and place their end one above another. Then, place this configuration between the two electrodes. Now, set up the voltage, current and time in the spot welding machine to join the two sheets (Fig. 3.7).

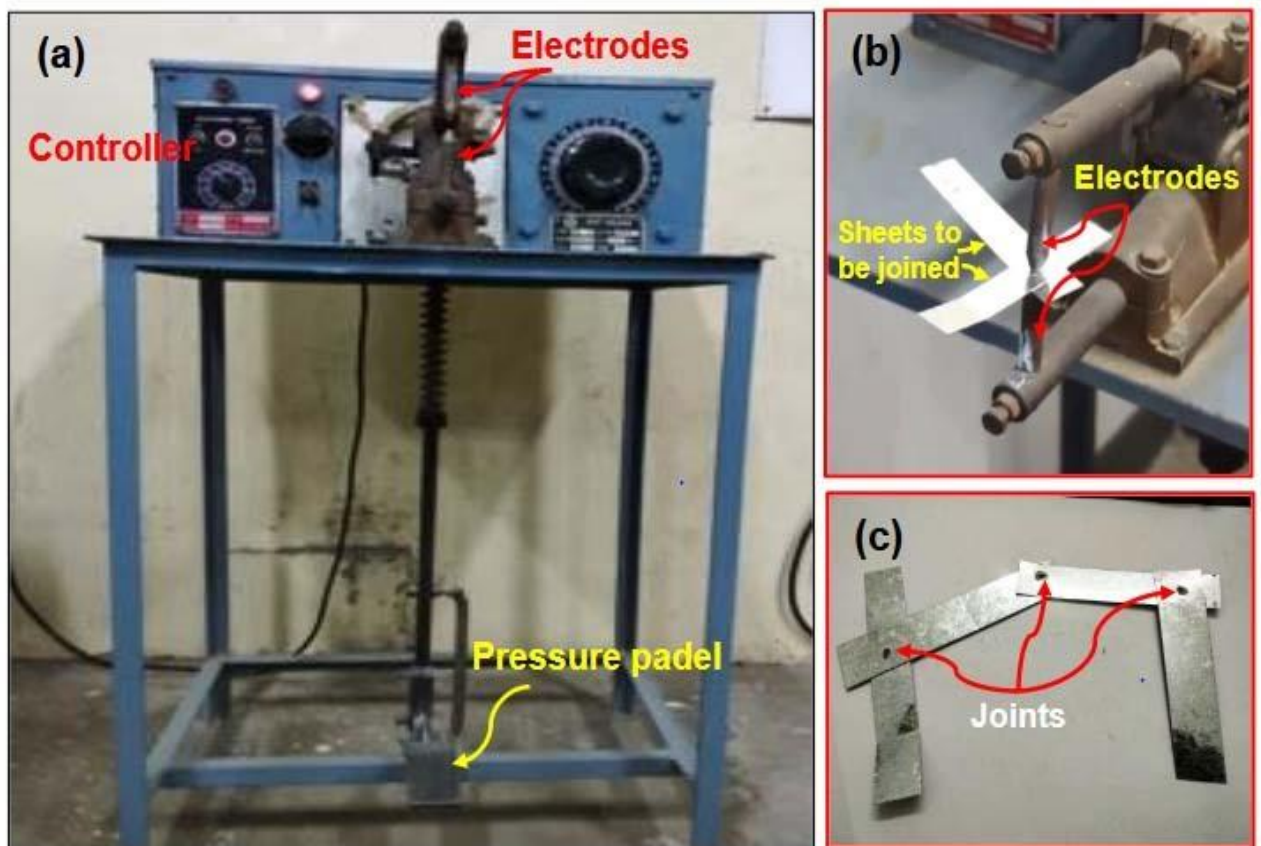
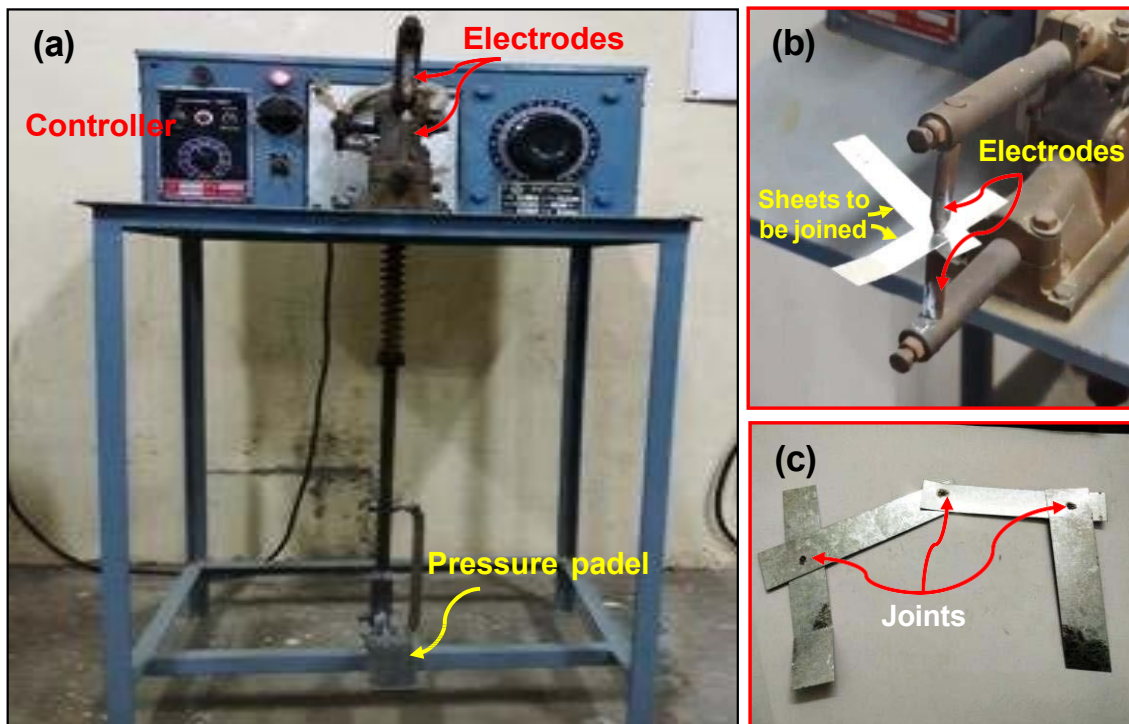


Fig. 3.7. Spot welding (a) setup and (b) process. (c) sheets after spot welding.

Fig. 3.7. Spot welding (a) setup and (b) process. (c) sheets after spot welding.

Experiment 4 (E4)

Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication

(A) Theory

Introduction to Brazing Process: In this process, metal parts to be joined are heated to a temperature below the melting point of the parts but sufficient to melt the filler material, which is used to fill the gap at the joint and establish a bond between the edges. The filler metal is drawn through the joint through the capillary action to create a joint between the two pieces (Fig. 4.1).

In this process, the base metal does not melt, and hence the metallurgy of the base metal is not disturbed much. This also suggests that the joints made by this process are not as strong as those made by welding processes. The advantage of brazing is that it can establish a joint between two dissimilar metals, through a proper choice of filler material. Unlike in welding, the filler rod differs widely in composition from the parent material(s) in brazing. Gas (oxy-acetylene mixture) is usually used for brazing.

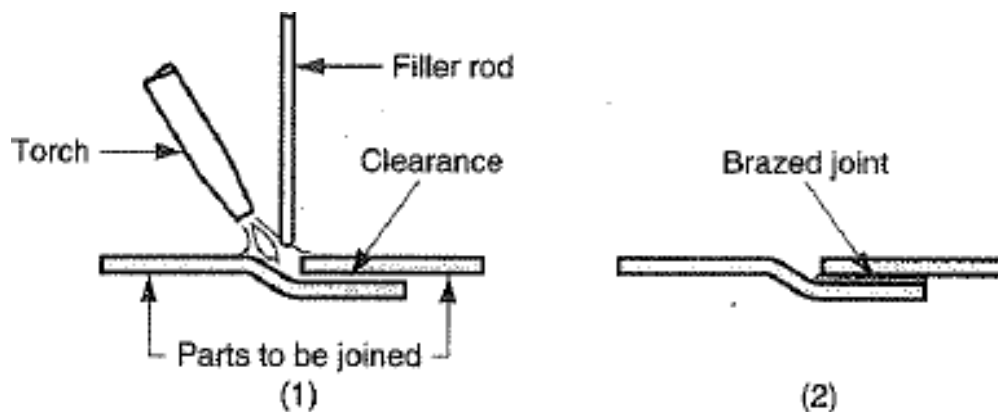


Fig. 4.1. (1) Schematic of a brazing process and (2) brazed joint.

(B) Exercise

The exercise will consist of making a joint by brazing process (Part - 1) and object fabrication (Part 2). Demonstration of drilling/cutting/grinding (Part - 3).

Part-1: Brazing process

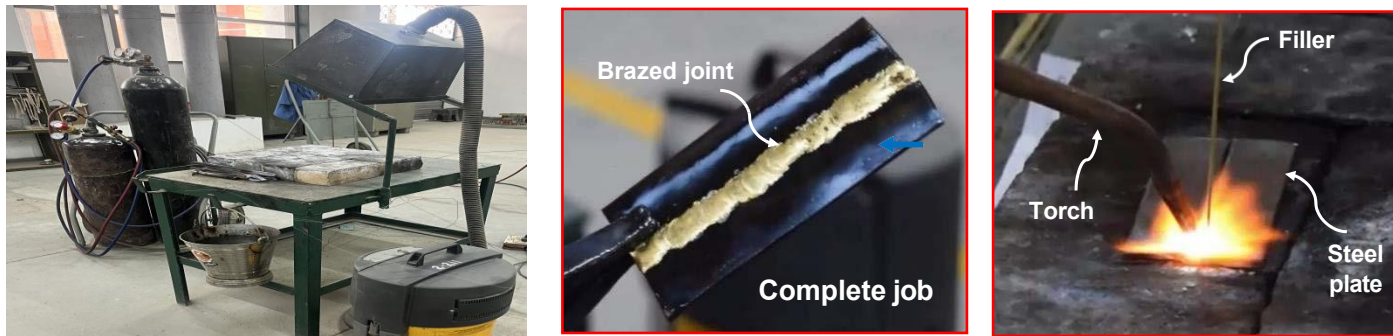
Objectives: To prepare a butt joint with mild steel strips using brazing technique.

Tools and Equipment: Oxy-Acetylene set, wire brush, tongs, safety gloves, safety goggles etc.

Materials: Brazing wire, flux, mild steel strips (100 x 25 x 3 mm)

Procedure:

- Clean the mild steel sheets by removing the oxide layer and flatten it.



• **Fig. 4.2.** Overall process of brazing process

Keep the metal sheet in lap position (see Fig. 4.2).

- Start the brazing process and maintain proper speed and feed.
- After the brazing process is complete, clean the joint using brush.

The overall brazing process is shown in Fig. 4.2.

Part - 2: Object fabrication

Objectives: To prepare a (1) spring, (2) square frame and (3) ring.

Tools and Equipment: Anvil, lightweight ball peen hammer, hand hacksaw, bench vice, pliers, supporting rod and file, etc.

Materials: Mild steel rod (4 mm), Copper coated steel wire (0.8 mm or 1.2 mm).

Procedure:

- 1) To make a spring from copper coated steel wire (0.8 mm or 1.2 mm) using bench vice and supporting rod (12 mm).
- 2) Making a ring from mild steel rod (4.0 mm) using bench vice and supporting rod (40 mm).
- 3) Bend a mild steel rod (4.0 mm) in a square frame using bench vice and hammer.



Fig. 4.3. Fabricated spring, ring and square frame.

Part - 3: Demonstration of drilling, cutting and grinding

Objectives: To learn (1) making hole on a plate, (2) cutting a pipe and (3) surface grinding.

Tools and Equipment: Drilling machine, grinding machine, cutting machine.

Materials: Mild steel pipe (25 mm), mild steel flat (5 mm)

Procedure:

- 1) Use drilling machine (Fig. 4.4a) and make a hole in 5 mm mild steel flat.
- 2) Cut a mild steel pipe using DoALL horizontal cutting machine (Fig. 4.4b).
- 3) Grind the surface of a steel flat (5 mm) using grinding machine (Fig. 4.4c).



Fig. 4.4. Equipment available in TA211 lab (a) Drilling machine, (b) Horizontal DoALL cutting machine, and (c) Grinding machine.

Experiment 5 (E5)

Demonstration of Rolling, Swaging, Injection Moulding and Tape Casting

(A) Rolling

Introduction to Rolling: Rolling is the process involving plastic deformation of metals by squeezing action as it passes between a pair of rotating rolls. To control the relative positioning of rolls, a roll positioning system is employed on the mill stand, generally through hydraulic pressure. Different configurations of high mills are shown in Fig. 5.1a. The most common rolling mill is the 2-high rolling mill and 4-high rolling mill. In 2-high mill, two rolls are usually mounted horizontally in bearings at their ends and vertically above each other rotating in opposite direction. The rolls may be driven through couplings at their ends by spindles, which are coupled, to pinions (or gears), which transmit the power from the electric motor. The rolling mills could be either reversing or non-reversing type. In the reversing type, the direction of motion of the rolls can be reversed, and therefore the work can be fed into the mill from both sides by reversing the direction of rotation of rolls (Typically used in industries). The rolling process may be carried out either at room temperature or at higher temperature. Depending on temperature, the rolling process is called cold rolling, warm rolling and hot rolling. Rolling below recrystallization temperature (generally $0.3 T_m$, where T is in Kelvin) and above $0.6 T_m$ (in Kelvin) is called cold rolling and hot rolling, respectively. Rolling between $0.3 T_m$ to $0.6 T_m$ will come under warm rolling.

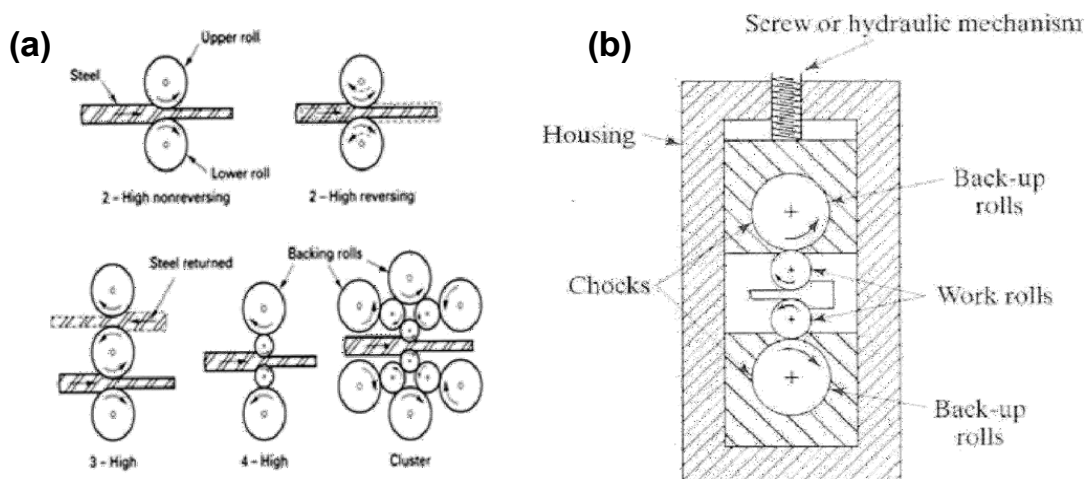


Fig. 5.1. (a) Different configurations of rolling mills and (b) configuration of 4-high mill.

Objectives: Understanding the rolling process by reducing the thickness of aluminum.

Equipment and Materials: Rolling machine and aluminium sheet.

Observations: Report the various parameters given in the Table below the rolling of the given metal.

- ✓ Roll Diameter:
- ✓ Roll Speed:
- ✓ Lubrication used:

S No.	Initial Dimensions			Lubricant conditions	Roll gap	Final dimensions			Draft	Spring back
	L	W	T			L	W	T		
1										
2										

Fig. 5.2 shows the different parts of a rolling machine along with the sheet before and after rolling process.



Fig. 5.2. (a) 2-high rolling machine and (b, c) sample before and after rolling, respectively.

(B) Swaging

Introduction to Swaging: Rotary swaging is a high-performance forging process for hollow and solid parts with an elongated axis. Rotary swaging is used in the manufacturing of various parts for the automotive, aerospace and defense industries. For example, swaging is used in the production of pipes, hollow steering columns, drive and guide shafts as well as various fasteners. The process of rotary swaging utilizes two to six dies for radial forming while the workpiece is moved axially

forward and/or backward. The dies may perform up to 10000 strokes per minute and their displacement might be as little as 0.2 mm per stroke.

Rotary swaging significantly reduces the production cost of such components as compared to machining processes and also increases strength properties of the components. Rotary swaging allows higher degrees of deformation compared to conventional metal forming processes without the need for intermediate annealing, consequently reducing the number of operations required for manufacturing finished parts. Furthermore, the overall time is also less.

Objective: To reduce the diameter of a solid rod through the swaging process.

Equipment and Materials: Swaging machine, Aluminum rod etc.

Fig. 5.3 shows the different parts of the swaging machine along with the rod before and after swaging process.

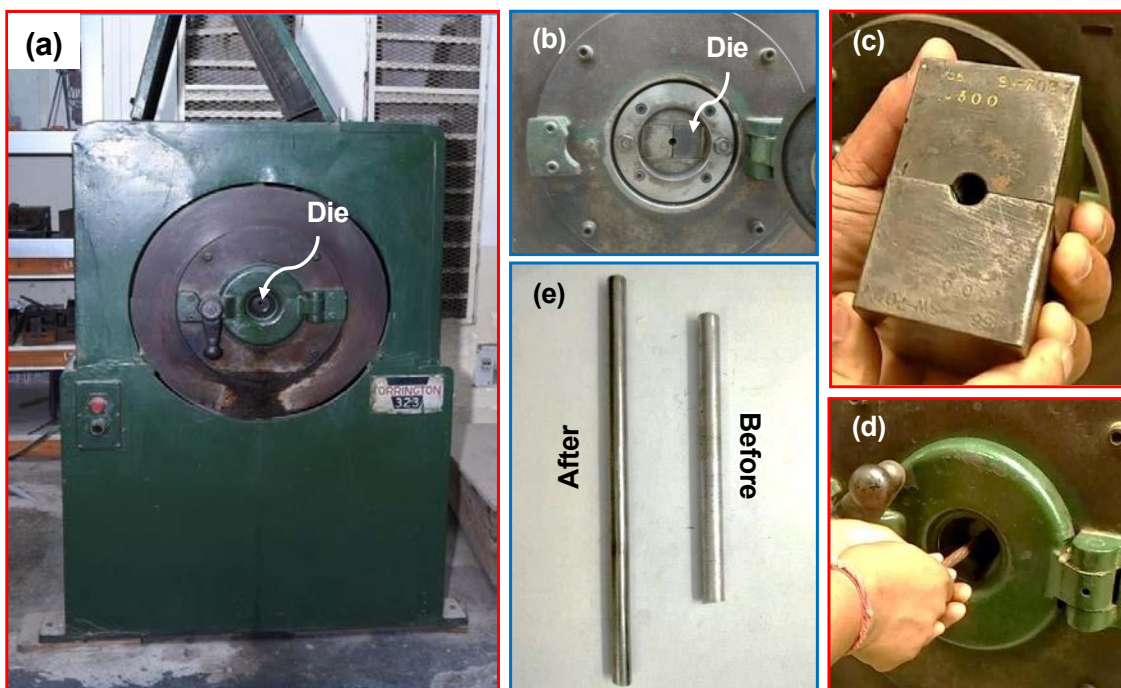


Fig. 5.3. (a) Swaging machine, (b, c) magnified versions of die, (d) swaging process, and (e) sample before and after swaging process.

(C) Plastic Injection Moulding

Introduction to Plastic Injection Moulding: In injection moulding, objects are formed by forcing molten plastic material under pressure into a mold where it is cooled, solidified and subsequently released by opening the two halves of the mold. Materials such as polystyrene, nylon,

polypropylene and polythene can be used in a process called injection molding. These are thermoplastics, i.e., when they are heated and pressured in a mould they can be formed into different shapes. Injection molding is used for the formation of intricate plastic parts with excellent dimensional accuracy. A large number of items associated with our daily life are produced by way of injection molding. Typical product categories include housewares, toys, automotive parts, furniture, rigid packaging items, appliances, medical disposable syringes, etc. Look around you and you will probably see dozens of injection molded parts in your wallet, kitchen, car and office. Through a schematic, the main parts of injection molding are as shown in Fig. 5.4.

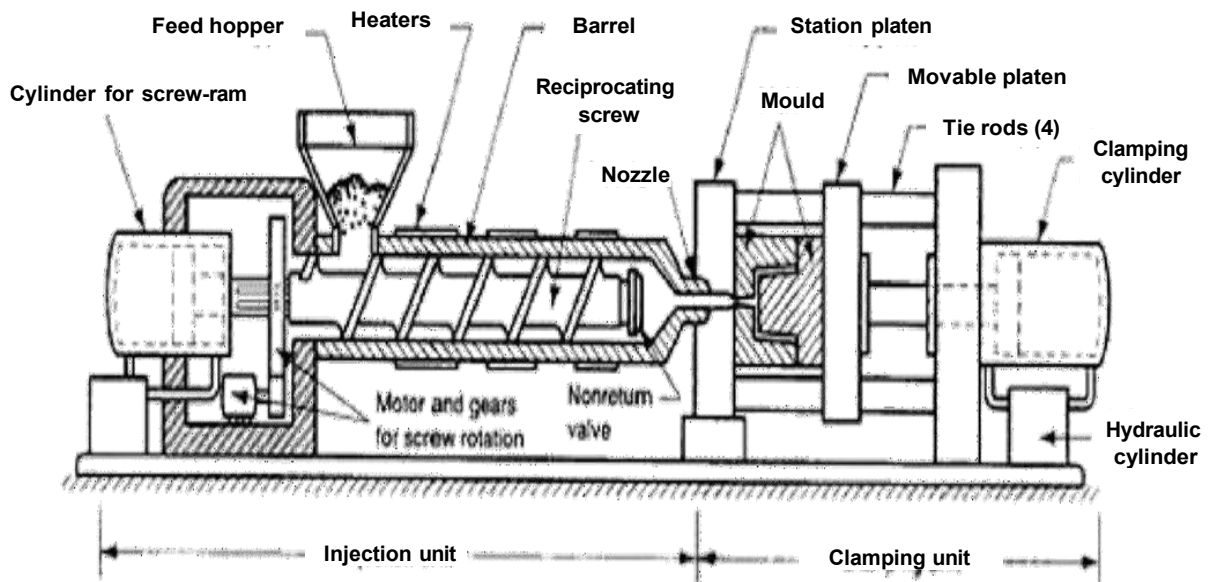


Fig. 5.4. Schematic of plastic injection molding.

Some of the advantages of plastic injection moulding include fast production, lower rejection rate, accuracy, choice of desired surface finish, and versatility in processing different raw materials. A few disadvantages include the high initial tooling cost and high running cost.

Objective: To fabricate tensile samples using plastic injection moulding process.

Equipment and Materials: plastic injection moulding machine, polypropylene.

Procedure: The process cycle for injection molding is very short, and consists of the following four stages:

1. **Clamping** - Prior to the injection of the material into the mold, the two halves of the mold must first be securely closed by the clamping unit. Each half of the mold is attached to the injection molding machine and one half is allowed to slide.

2. **Injection** - The raw plastic material, usually in the form of pellets, is fed into the injection molding machine, and advanced towards the mold by the injection unit. During this process, the material is melted by heat and pressure. The molten plastic is then injected into the mold very quickly and the buildup of pressure packs and holds the material.
3. **Cooling** - The molten plastic that is inside the mold begins to cool as soon as it makes contact with the interior mold surfaces. As the plastic cools, it will solidify into the shape of the desired part. However, during cooling some shrinkage of the part may occur.
4. **Ejection** - After sufficient time has passed, the cooled part needs to be removed. In some machines, the process of ejection is automatic.

During cooling, the material in the channels of the mold will solidify attached to the part. This excess material, along with any flash that has occurred, must be trimmed from the part, typically by using cutters. The set-up of injection moulding along with dies and fabricated samples are shown in Fig. 5.5.

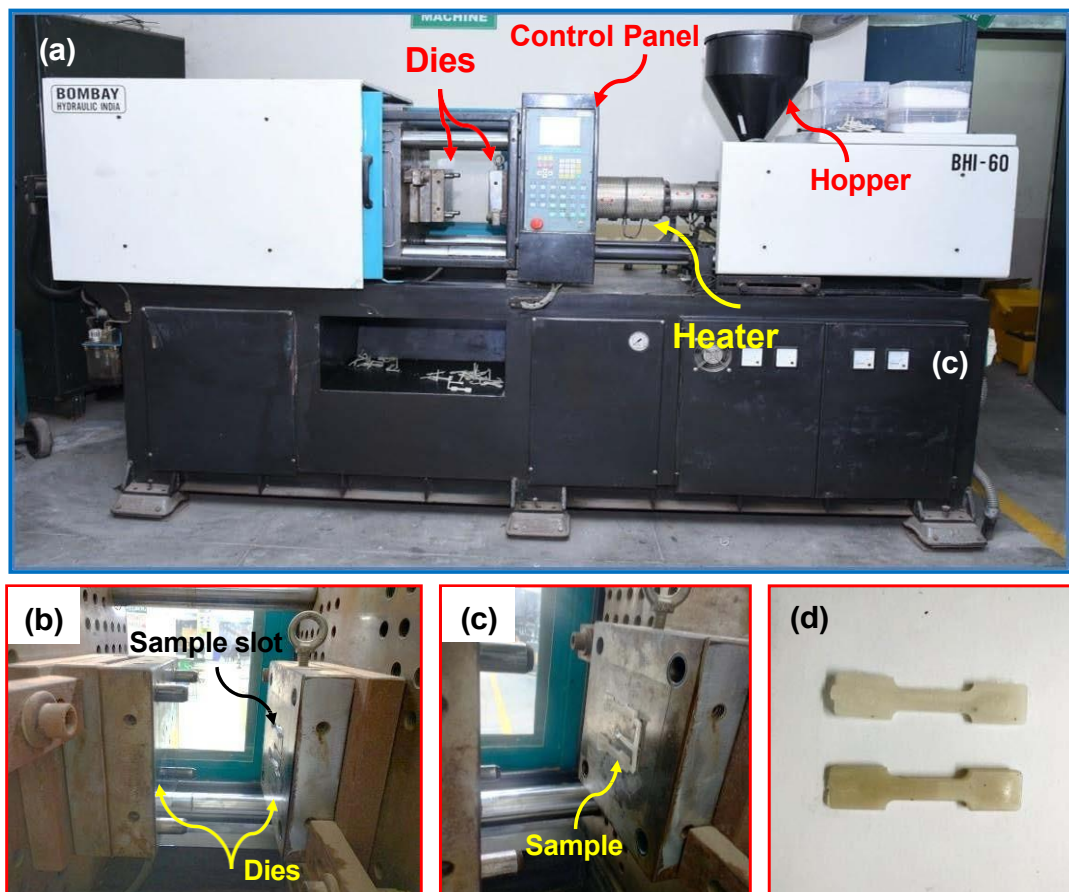


Fig. 5.5. (a) Injection moulding machine, (b) dies, (c, d) fabricated samples.

(D) Tape Casting

Introduction to Tape Casting: Tape casting is preferably used for the production of large, thin ceramic layers (wide-flat layers). The ceramic raw powders and a dispersing agent are dispersed in a liquid and mixed with an organic binder system. The homogenous, evacuated slurry is given into the slurry reservoir and flows under an exactly adjusted doctor-blade onto the carrier tape. During the subsequent drying process, the dispersing liquid is evaporated where the tape thickness reduces. Casted tapes range commonly between 20 μm and 1 mm in thickness. Multi-layer tapes with different powders in each layer are prepared by repeating the casting steps or by laminating the single-layer tapes. In this demonstration, Colgate toothpaste will be utilized instead of ceramics etc. (Fig. 5.6)

Tape casting can be used to develop green tapes of ceramic, metal or glass powders. Some of the examples include development of two-layered silicon carbide tapes with coarse-porous support and fine-porous membrane layer and then structuring and joining of these tapes to produce flat multi-channel elements for membrane filtration, thin tapes ($\leq 50 \mu\text{m}$) for diffusion joining of non -oxide ceramic parts, structured Si-SiC tapes for thermoelectric applications (e.g., heating elements) and development of oxide ceramic tapes for different applications (e.g., dense zirconia, porous alumina).

Objective: To create a layer using tape casting process.

Equipment and Materials: Tape casting machine, Colgate etc.

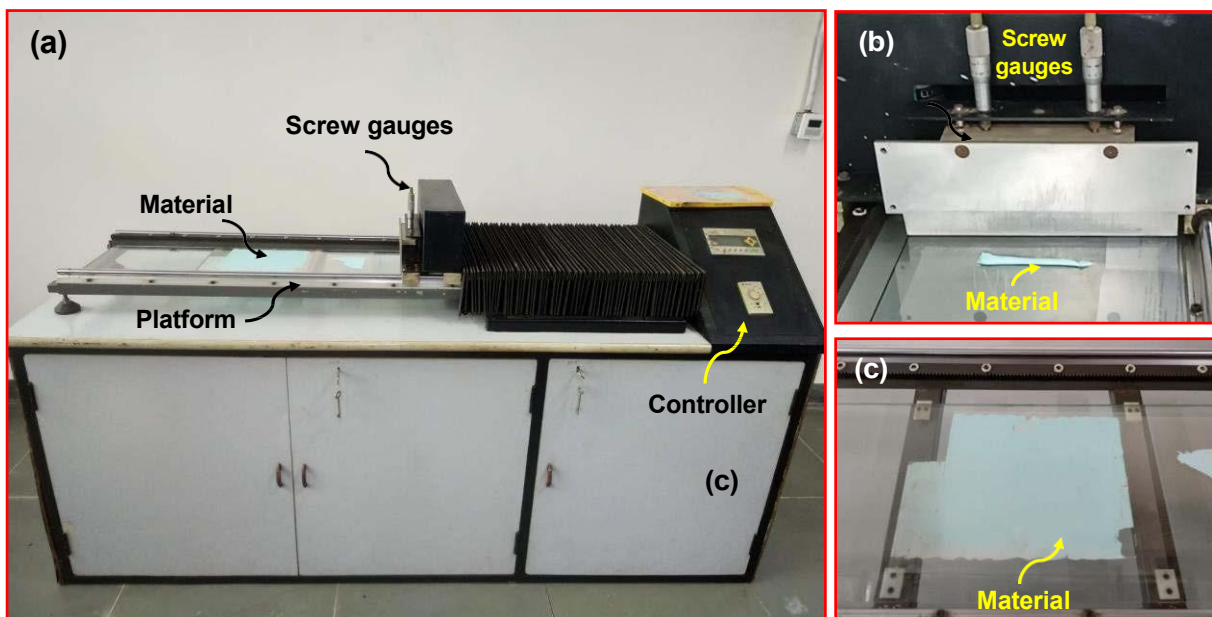


Fig. 5.6. (a) Tape casting setup. (b) and (c) are images before and after tape casting process.